

Assignment-5

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Tutorial sheet -10

Section - A

1) Define digital transmission?

Ans Digital transmission is the process of sending binary data (0s and 1s) over a network or communication medium. It's widely used for transmitting information in a format that computers and digital devices can interpret.

2) What is wide area network (WAN)?

Ans WAN connects multiple local area network (LAN's) over a large geographic area. It enables the exchange of data and communication between distant locations.

3) Write 3 names of routing algorithm?

Ans → Dijkstra's algorithm :- find the shortest path in graph
→ Bellman-Ford algorithm :- Determines the shortest path suitable for distributed networks.
→ OSPF (Open shortest path first) :- A dynamic routing protocol used in IP.

4) Define the term congestion control?

Ans Congestion control manages network traffic to prevent congestion, ensuring smooth data flow and preventing performance issues.

5) What is Domain name system (DNS)?

Ans The DNS translates human readable domain names into numerical IP addresses, simplifying how users access websites and resources on the internet.

Section - B

1) Explain flow control techniques in data link layer.

Ans 1) stop and wait :- Simplest technique where the sender sends one frame and waits for an acknowledgment before sending the next. Inefficient for high latency networks as it underutilizes the link.

2) Sliding window :- Allows multiple frames to be in transit simultaneously. The sender can send a "window" of frames before requiring an acknowledgment.

3) Selective repeat :- A variation of sliding window where only the frames with errors are retransmitted, improving efficiency. Requires more complex error control mechanisms.

4) Automatic Repeat reQuest (ARQ) :- Detects and requests retransmission of lost or corrupted frames automatically. Commonly used in modern data link layer protocols.

84 Flow Control with Credits:- Receiver issues credits to the sender, specifying the number of frames it can accept. Enhances efficiency by avoiding overloading the receiver.

85 Windows Size Adjustment:- Dynamic adjustment of the window size based on network conditions. Optimizes data transfer by adapting to varying latency and bandwidth.

87 Buffering and Rate Limiting:- Buffers at sender and receiver sides help smooth out variations in data arrival rates. Rate limiting ensures that the sender doesn't overwhelm the receiver.

27 Explain Discuss the 3 main divisions of domain name space?

Ans The domain name space is organized hierarchically and it comprises three main divisions:-

1) Top-level Domain (TLD):- The highest level in the domain name hierarchy

→ Eg:- .com, .org, .net, .gov, .edu & country code TLD's like .uk, .jp.

→ TLD's categorize domain names based on their general purpose, such as commercial and organizational, network infrastructure, government, educational and geographical issues.

2) Second-level Domain (SLD):- Directly beneath the TLD.

→ Eg:- in "example.com" example is the SLD

→ SLD's often represent organizations, entities or specific brand users can register domain names under a chosen TLD & create unique SLD for their websites.

3) Subdomain: - A division beneath the SLD

→ Eg:- In "blog.example.com," "blog" is a subdomain

→ It allow organizations to further organize their domain space hierarchically. They are used for specific sections or services within a domain.

For eg:- a company might have subdomains for different departments or services like "sales.example.com" or "support.example.com".

3) What do you mean by computer networks? Classify the computer network and explain them in brief?

Ans A computer network is a collection of interconnected computers or computing devices that communicate and share resources with each other. Networks can be as simple as two computers connected in a home setting or as complex as a global network like the Internet. The primary purpose of computer networks is to facilitate the exchange of information and resources among connected devices.

Classification of computer Networks

1) Based on Geographical Scope:-

→ LAN

→ limited to a small geographic area, such as a single building, office or campus

→ High data transfer rates, low latency -

→ Office or school network.

→ WAN

⇒ Spans a larger geographical area, often connecting LANs across countries and continents.

⇒ Lower data transfer rates compared to LAN higher latency.

⇒ The Internet, connecting LANs globally.

→ MAN

⇒ Covers a city or a large campus.

⇒ Intermediate size and data transfer rates b/w LAN & WAN.

⇒ City-wide network connecting multiple buildings.

2) Based on Functional Relationships:-

→ Client-Server Network

⇒ Clients request services or resources from servers.

⇒ Websites, database servers, file servers.

→ Peer-to-Peer Networks

⇒ All devices have equal status and can share resources without a central server.

⇒ Home networks, file sharing network.

Section - C

1) If IP address is 23.192.157.234 and subnet mask is 255.255.0.0 then find:

a) No. of hosts

b) No. of Subnet

c) Subnet ID of each Subnet.

Ans IP address = 23.192.157.234

Subnet Mask = 255.128.0.0

Binary representation of:-

IP address = 00010111.11000000.10011101.11101010

Subnet Mask = 11111111.10000000.00000000.00000000

a) The subnet mask 255.128.0.0 corresponds to a /9 subnet in CIDR notation

Therefore, no. of host bits is $32 - 9 = 23$ bits

The formula for calculating the no. of hosts is $2^{(\text{no. of host bits})} - 2$

So, $2^{23} - 2 = 8,388,606$ hosts

b) ~~The subnet mask 255.128.0.0~~

The formula for calculating the no. of subnets is $2^{(\text{no. of subnet bits})}$

So, $2^{23} = 8,388,608$ subnets

c) To find the subnet ID of each subnet, you need to determine the increment value based on the subnet mask.

The increment value is $2^{(32 - \text{subnet mask length})}$

In this case 2^{23} is the increment value.

The first subnet ID is 23.0.0.0, the second subnet ID is 23.128.0.0 & so on.

2) Explain slow start algorithm in detail.
 Slow start is a congestion control algorithm used in network protocols like TCP (Transmission Control Protocol) to control the rate at which new connections are established. Here's an

1) Initialization:-

- When a connection is established or re-established, the sender starts in the slow start state.
- The sender begins by sending a small no. of packets into the network.

2) Exponential Growth

- The sender exponentially increases the no. of packets it sends for each round-trip time.
- After each acknowledgment is received for sent packets, the sender doubles the sending rate.
- This rapid growth is the 'exponential' part of slow start.

3) Congestion Window

Maintains a congestion window (cwnd) representing unacknowledged packets. It is dynamically adjusted during slow start.

4) Threshold:

It continues until a predefined threshold is reached. It is often set by the network or the sender based on factors like network conditions & congestion.

5) Transition to Congestion Avoidance

Once the congestion window reaches the threshold, the algorithm transitions from slow start to the congestion avoidance state.

6) Congestion Avoidance

In it, the sender increases the congestion window linearly rather than exponentially. The rate of increase is slower to avoid potential network congestion.

Retransmission on Timeout:-

If a timeout occurs, the sender assumes network congestion and reduces the congestion window to a smaller value, and enters slow start again.

8% Adaptive Behaviour :-

It is adaptive, it allows the sender to quickly explore the available network capacity without causing congestion. If the network can handle the increased rate, the sender accelerates its transmission speed.

9% Dynamic Adjustment :-

It dynamically adjusts to the network conditions by starting with a small window size and increasing it gradually. This prevents the sender from overwhelming the network with too much traffic at the beginning of a connection.

10% Connection Stability :-

Slow start helps in stabilizing the connection and finding an optimal transmission rate without causing congestion.

3% Explain AIMD algorithm in detail.

Ans. Additive Increase, Multiplicative decrease algorithm.

1% Initialization :-

- Start with a congestion window size (cwnd) of 1 or small value.
- Set a slow start threshold initially.

2% Additive Increase

- During each successful round-time, increase the congestion window by a fixed amount.
- This phase allows the sender to gradually probe the network's capacity.

3% Congestion Avoidance :-

- Once the cwnd surpasses the slow start threshold, switch to congestion avoidance mode.
- In congestion avoidance, increase the congestion window linearly by 1 for each round trip time.

4x Multiplicative decrease:-

- If packet loss is detected, cut the congestion window size in half.
- Set the slow start threshold to half of the current congestion window size

5x Fast Recovery

- After detecting packet loss, enter Fast Recovery.
- During Fast Recovery, increase the congestion window linearly for each acknowledgment.

6x Adaptive Behavior

- AIMD adapts to network conditions by probing capacity in the additive increase phase and adjusting the congestion window size in response to congestion events.

7x Stability & Fairness

- AIMD helps maintain network stability by avoiding rapid increases in congestion and reacting quickly to packet loss.
- Provides fair sharing of network resources among competing connections.

8x Continuation

- Continue the AIMD cycle, iteratively adjusting the congestion window based on network conditions.
- This adaptive behavior contributes to effective congestion control in TCP connections.